Pengcheng (David) Ye

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Education

Imperial College London – *PhD in Mathematics (2026 – Now)*

- *Supervision*: Dr. Yanbo Tang and Dr. Riccardo Passeggeri
- *Research Directions*: Optimal Transport, Interacting Particle Systems, Mean Field Theory
- *Funding*: Fully funded by the Roth Scholarship

Imperial College London – *BSc in Mathematics (2023 – 2026)*

- *Final Award*: First Class Honours (86.48)
- *Final Year Modules*: Probability Theory 1, Functional Analysis, Probability Theory 2, Statistical Mechanics, Stochastic Differential Equations and Diffusion Processes, Fourier Analysis and Theory of Distributions, Probabilistic Generative Models, Methods for Data Science
- *Second Year Project*: Viterbi Algorithm and its Applications
- *First Year Project*: Seismographic Modelling via Hawkes Processes

American International School of Budapest – *IB Diploma (2021 – 2023)*

- *HL Courses*: Computer Science (7), Mathematics AA (7), Physics (7)
- *SL Courses*: Economics (7), French (7), English Language and Literature (6)
- *Extended Essay in Mathematics*: An Investigation into the Constructability of Regular Polygons

Work Experience

Acquiring Risk and AI Intern – *Revolut, London, UK (2026)*

- Developed smart retry logic for subscription payments to decrease involuntary churn.
- Engineered SEO-based data features for acquiring merchant fraud detection models.
- Scoped a predictive buyer fraud model, extracting key features from acquiring and issuing dispute (TC40 and chargeback) data.

Trading Intern – *IMC Trading, Amsterdam, Netherlands (2025)*

- Studied basic options theory and practised trading intuition through weekly training and market simulations.
- Analysed market microstructure and order flow dynamics on India equity options desk.
- Developed quoting strategy targeting low base-price options based on empirical findings.

Research Experience

Undergraduate Research Opportunities Programme – *Imperial College London (2026)*

- Formulated optimal transport theory between random measures, motivated by empirical measure and Bayesian analysis settings. Derived statistical convergence results under this formulation, with concrete applications in generative modelling.
- *Supervision*: Dr. Riccardo Passeggeri
- *Funding*: Imperial College UROP Award (£5,088) and London Mathematical Society Undergraduate Research Bursary (£2,680)

Imperial College – NTU Singapore Statistics Research Project – *Imperial College London (2025)*

- Investigated the use of ordinal regression for fragility curve modelling, over traditional nominal methods. Presented statistical improvements for natural hazard loss assessments in collaboration with peers from Imperial and NTU.
- *Supervision*: Dr. Ioanna Papatsouma, Dr. Christopher Hallsworth and Dr. Michele Nguyen
- *Funding*: NTU-Imperial Collaboration Fund (£1,320)

Publications & Preprints

L^2 over Wasserstein: Statistical Analysis for Optimal Transport – *Preprint (2026)*

- Riccardo Passeggeri, Rohan M. Shenoy, Pengcheng Ye
- arXiv:2605.21365, Statistics Theory (math.ST) (arXiv)

Teaching

Graduate Teaching Assistant – *Imperial College London · Department of Mathematics (2026 – Now)*

- MATH50001 Analysis 2 (Autumn Term 2026)

Undergraduate Peer Tutor – *Imperial College London · Department of Mathematics (2025 – 2026)*

- All first-year mathematics modules (MATH40001–7)

Maths Circle Mentor – *Axiom Maths · Woodhouse College (2024 – 2025)*

- Delivered interesting maths topics outside standard curriculum through group discussion and problem-solving.

Outreach

Directed Reading Programme Leader – *Imperial College London · Department of Mathematics (2025 – Now)*

- *Group*: Mean Field Theory and Applications (2026 – Now)
- *Group*: Statistical Optimal Transport (2025 – 2026)

A-level Further Maths Tutor – *Imperial College London · Imperial Outreach (2024 – 2025)*

- Supported a group of Y13 students with Further Maths and STEP problems in preparation for university.

Data Science and AI Academic Mentor – *Imperial College London · Global Summer School (2024)*

- Mentored a group of high school students in Data Science through Python workshops on real-world datasets.

Awards

Peer Tutor of the Year Award – *Imperial College London (2026)*

Exceptional Contribution Award – *Imperial College London (2026)*

Dean's List – *Imperial College London (2023 – 2026)*

- Years 1, 2 and 3

Skills & Interests

Language: English, Mandarin, Hungarian, French (B1)
Coding Language: Python, Java, R
Hobbies: Chess, Catan, Running, Racket Sports